

CPU:

Architecture: x86\_64  
CPU(s): 80  
Vendor ID: GenuineIntel  
Model name: Intel(R) Xeon(R) Gold 6248 CPU @ 2.50GHz  
Caches (sum of all):  
L1d: 1.3 MiB (40 instances)  
L1i: 1.3 MiB (40 instances)  
L2: 40 MiB (40 instances)  
L3: 55 MiB (2 instances)  
NUMA:  
NUMA node0 CPU(s): 0-19,40-59  
NUMA node1 CPU(s): 20-39,60-79

Название сервера:

ProLiant XL270d Gen10

Доступные numa nodes:

available: 2 nodes (0-1)  
node 0 size: 385636 MB  
node 1 size: 387008 MB

Система: Ubuntu 22.04.5 LTS

The graph illustrates the performance of three different methods in terms of the number of iterations required as the number of nodes increases from 0 to 40. The Linear method (yellow line) shows a consistent, linear increase in iterations, reaching 40 iterations at 40 nodes. The 20000x20000 method (blue line) and the 40000x40000 method (red line) both show a sharp increase in iterations up to 20 nodes, after which they plateau. The 40000x40000 method generally requires slightly more iterations than the 20000x20000 method for 20 or more nodes.

| Number of Nodes | 20000x20000 (Iterations) | 40000x40000 (Iterations) | Linear (Iterations) |
|-----------------|--------------------------|--------------------------|---------------------|
| 0               | 2                        | 2                        | 2                   |
| 10              | 9                        | 9                        | 10                  |
| 20              | 16                       | 16                       | 20                  |
| 30              | 16                       | 16                       | 30                  |
| 40              | 16.5                     | 15.5                     | 40                  |